EXTENSION 5F Multiplying and dividing algebraic fractions

Learning intentions

- To understand that the rules of multiplying and dividing fractions extend to algebraic fractions
- To be able to multiply algebraic fractions and simplify the result
- To be able to divide algebraic fractions and simplify the result

Key Ideas

- To multiply two algebraic fractions, multiply the **numerators** and the **denominators** separately. Then cancel any common factors in the numerator and the denominator.
  For example:
  $$\frac{2x}{5} \times \frac{10y}{3} = \frac{\cancel{20}^{4}xy}{\cancel{15}^{3}} = \frac{4xy}{3}$$
- The **reciprocal** of an algebraic fraction is formed by swapping the numerator and denominator.
  The reciprocal of $\frac{3b}{4}$ is $\frac{4}{3b}$.
- To divide algebraic fractions, take the reciprocal of the second fraction and then multiply.
  For example:
  $$\frac{2a}{5} \div \frac{3b}{4} = \frac{2a}{5} \times \frac{4}{3b} = \frac{8a}{15b}$$

Example 10 | Multiplying algebraic fractions

Simplify the following products.

a $\frac{2a}{5} \times \frac{3b}{7}$

$= \frac{6ab}{35}$

b $\frac{4x}{15} \times \frac{3y}{2}$

$= \frac{\cancel{12}^{2}xy}{{}^{5}\cancel{30}}$

$= \frac{2xy}{5}$

Example 11 | Dividing algebraic fractions

Simplify the following divisions.

a $\frac{3a}{8} \div \frac{b}{5}$

$= \frac{3a}{8} \times \frac{5}{b}$

$= \frac{15a}{8b}$

b $\frac{u}{4} \div \frac{15p}{2}$

$= \frac{u}{4} \times \frac{2}{15p}$

$= \frac{\cancel{2}^{1}u}{_{30}\cancel{60}p}$

$= \frac{u}{30p}$